

Outputs

The below output shows ingestion of healthcare pdf documents to FAISS vector store using the file “ingest.py” file.

I have used the “check_ingestion.py” file to validate whether the documents were ingested properly or not.

I ran the chatbot using “healthcarebot.py” file then asked the questions that are relevant to the documents that I have ingested.

```
Terminal x healthcarebot.py x ./work/work/RAG Project/ingest.py x
labsuser@jupyter:~$ cd work
labsuser@jupyter:~/work$ cd "RAG Project"
labsuser@jupyter:~/work/RAG Project$ python ingest.py
/usr/local/lib/python3.10/site-packages/transformers/utils/generic.py:441: FutureWarning: `torch.utils._pytree._register_pytree_node` is deprecated. Please use `torch.utils._pytree.register_pytree_node` instead.
  _torch_pytree._register_pytree_node(
/usr/local/lib/python3.10/site-packages/pypdf/_crypt_providers/_cryptography.py:32: CryptographyDeprecationWarning: ARC4 has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.ARC4 and will be removed from cryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  from cryptography.hazmat.primitives.ciphers.algorithms import AES, ARC4
loaded 95 pages from PDFs
created 196 text chunks
/usr/local/lib/python3.10/site-packages/trio_core/_multierror.py:511: RuntimeWarning: You seem to already have a custom sys.excepthook handler installed. I'll skip installing Trio's custom handler, but this means MultiErrors will not show full tracebacks.
  warnings.warn(
Healthcare Knowledge base successfully indexed
labsuser@jupyter:~/work/RAG Project$ python check_ingestion.py
/usr/local/lib/python3.10/site-packages/transformers/utils/generic.py:441: FutureWarning: `torch.utils._pytree._register_pytree_node` is deprecated. Please use `torch.utils._pytree.register_pytree_node` instead.
  _torch_pytree._register_pytree_node(
/usr/local/lib/python3.10/site-packages/pypdf/_crypt_providers/_cryptography.py:32: CryptographyDeprecationWarning: ARC4 has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.ARC4 and will be removed from cryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  from cryptography.hazmat.primitives.ciphers.algorithms import AES, ARC4
PDF: Animal bites.pdf | Pages: 5 | Chunks: 13
PDF: Cardiovascular diseases.pdf | Pages: 4 | Chunks: 9
PDF: nutrition guide.pdf | Pages: 20 | Chunks: 26
PDF: Andhra Pradesh Arogyasri.pdf | Pages: 50 | Chunks: 115
PDF: DIABETES.pdf | Pages: 4 | Chunks: 7
PDF: HIV and AIDS.pdf | Pages: 4 | Chunks: 9
PDF: TXH Healthcare queries.pdf | Pages: 3 | Chunks: 5
PDF: Vaccines and immunization.pdf | Pages: 5 | Chunks: 12
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Total PDFs: 8
Total Pages: 95
Total Chunks: 196
labsuser@jupyter:~/work/RAG Project$ python healthcarebot.py
/usr/local/lib/python3.10/site-packages/transformers/utils/generic.py:441: FutureWarning: `torch.utils._pytree._register_pytree_node` is deprecated. Please use `torch.utils._pytree.register_pytree_node` instead.
  _torch_pytree._register_pytree_node(
/voc/work/work/RAG Project/healthcarebot.py:16: LangChainDeprecationWarning: Please see the migration guide at: https://python.langchain.com/docs/versions/migrating_memory/
```

Healthcarebot.py

1. HIV and AIDS

Shows the output w.r.t HIV and AIDS pdf

```
labsuser@jupyter:~/work/RAG Project$ python healthcarebot.py
/usr/local/lib/python3.10/site-packages/transformers/utils/generic.py:441: FutureWarning: `torch.utils._pytree._register_pytree_node` is deprecated. Please use `torch.utils._pytree.register_pytree_node` instead.
  _torch_pytree._register_pytree_node(
/voc/work/work/RAG Project/healthcarebot.py:16: LangChainDeprecationWarning: Please see the migration guide at: https://python.langchain.com/docs/versions/migrating_memory/
  memory = ConversationBufferMemory(
/usr/local/lib/python3.10/site-packages/trio/_core/_multierror.py:511: RuntimeWarning: You seem to already have a custom sys.excepthook handler installed. I'll skip installing Trio's custom handler, but this means MultiErrors will not show full tracebacks.
  warnings.warn(

Healthcare Assistant (type 'exit' to quit)

User: Nutrition guidelines for diabetes

Healthcare Assistant: I'm not sure based on the available healthcare documents.
-----
User: can you tell me the symptoms of HIV?

Healthcare Assistant: The signs and symptoms of HIV vary depending on the stage of infection. In the first few weeks after being infected, some people may experience flu-like symptoms, which can include:

- Fever
- Headache
- Rash
- Sore throat

As the infection progresses and weakens the immune system, other signs and symptoms may develop, such as:

- Swollen lymph nodes
- Weight loss
- Fever
- Diarrhea
- Cough

Without treatment, people living with HIV can also develop severe illnesses, including tuberculosis (TB), cryptococcal meningitis, severe bacterial infections, and certain cancers like lymphomas and Kaposi's sarcoma (HIV and AIDS.pdf, Page 1).
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User: any remedies to get rid of HIV?

Healthcare Assistant: I'm not sure based on the available healthcare documents.
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User: is HIV preventable and how?

Healthcare Assistant: Yes, HIV is a preventable disease. The risk of HIV infection can be reduced by:

- Using a male or female condom during sex.
- Being tested for HIV and other sexually transmitted infections.
- Being circumcised if you are a man.
- Using harm reduction services if you inject and use drugs.

Additionally, pre-exposure prophylaxis (PrEP) is an option for HIV-negative people to reduce the risk of HIV acquisition. WHO recommends several PrEP methods, including:

- Oral tenofovir (TDF)-based PrEP
- Dapivirine vaginal ring
- Long-acting injectable cabotegravir
- Long-acting injectable lenacapavir

Access to testing and antiretroviral therapy (ART) is also crucial in preventing HIV transmission (HIV and AIDS.pdf, Page 3).
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User: █
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2. TXH Hospitals

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User: can i take a checkup for this in your hospital?

Healthcare Assistant: Yes, you can take a checkup at TX Hospitals. To book an appointment, you can visit our website or call our helpline numbers for assistance. It is recommended to schedule an appointment to minimize your waiting time (TXH Healthcare queries.pdf, Page 2).
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User: is ambulance facility available

Healthcare Assistant: Yes, we provide ambulance service for the pickup of patients from their residence or hospital to our hospital, and for dropping from our hospital to their residence. In case any service is not available at our hospital, we also provide ambulance service to a recommended facility of your choice (TXH Healthcare queries.pdf, Page 2).
```

3. Arogyasri

User: my mother has serious cardiac arrest, i want to take her to the hospital, does your hospital have any facility for that, and is there an arogyasri facility for this?

Healthcare Assistant: Yes, we provide ambulance service for the pickup of patients from their residence or hospital to our hospital. In case any service is not available at our hospital, we also provide ambulance service to a recommended facility of your choice (TXH Healthcare queries.pdf, Page 2).

Regarding the Arogyasri facility, the Dr. YSR Arogyasri Scheme covers various procedures, including those related to cardiac care. It provides end-to-end cashless services for eligible beneficiaries (Andhra Pradesh Arogyasri.pdf, Page 3). You may want to check if your mother qualifies for this scheme.

User: how much can be reimbursed using arogyasri

Healthcare Assistant: Under the Dr. YSR Arogyasri Scheme, the financial limit for reimbursement has been enhanced to Rs. 5.00 lakhs per family per annum for providing cashless treatment for Below Poverty Line (BPL) families (Andhra Pradesh Arogyasri.pdf, Page 3).

4. Vaccination and Immunization

User: does infants need vaccination? if yes what are they?

Healthcare Assistant: Yes, infants need vaccinations to protect them from various infectious diseases. Vaccination is one of the most effective ways to prevent diseases and has saved millions of lives. The specific vaccines recommended for infants typically include:

- Hepatitis B
- Diphtheria, Tetanus, and Pertussis (DTaP)
- Haemophilus influenzae type b (Hib)
- Polio (IPV)
- Pneumococcal conjugate vaccine (PCV)
- Measles, Mumps, and Rubella (MMR)
- Varicella (chickenpox)

These vaccines are usually administered according to a schedule recommended by health authorities to ensure optimal protection (Vaccines and immunization.pdf, Page 1).

5. Diabetes

User: what are the types of diabetes?

Healthcare Assistant: There are three main types of diabetes:

1. **Type 1 Diabetes**: This type is characterized by deficient insulin production and requires daily administration of insulin. It is often referred to as insulin-dependent or juvenile diabetes.
2. **Type 2 Diabetes**: This type affects how the body uses sugar (glucose) for energy. It stops the body from using insulin properly, which can lead to high levels of blood sugar if not treated. Type 2 diabetes is often preventable and is the most common form of diabetes.
3. **Gestational Diabetes**: This type occurs during pregnancy and is characterized by hyperglycemia with blood glucose values above normal but below those diagnostic of diabetes. Women with gestational diabetes are at an increased risk of complications during pregnancy and may also be at risk for developing type 2 diabetes in the future (DIABETES.pdf, Page 2).

6. Nutrition Guide

User: what food should diabetes patient should take as per nutrition guide?

Healthcare Assistant: Diabetes patients should focus on a healthy diet that includes:

- **Fruits and Vegetables**: Aim for at least 5 portions (400 g) of fruits and vegetables each day. This helps reduce the risk of noncommunicable diseases and ensures adequate dietary fiber intake.
- **Whole Grains**: Include whole grains such as unprocessed maize, millet, oats, wheat, and brown rice.
- **Legumes**: Incorporate legumes like lentils and beans into meals.
- **Nuts**: Include nuts as part of the diet.
- **Lean Proteins**: Consume 160 g of meat and beans daily, with red meat limited to 1-2 times per week and poultry 2-3 times per week.
- **Limit Sugars and Fats**: Reduce total energy intake from free sugars to around 12 teaspoons (50 g) or ideally less than 5% of total energy intake. Limit total fat intake to less than 30%, focusing on unsaturated fats (found in fish, avocado, nuts, and certain oils) and minimizing saturated and trans fats.
- **Salt Intake**: Limit salt to less than 5 g per day (equivalent to about 1 teaspoon) (nutrition guide.pdf, Page 11 and Page 14).

These dietary choices can help manage blood sugar levels and overall health for individuals with diabetes.

7. Cardiovascular diseases

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User: what are the riskfactors of heart attack

Healthcare Assistant: The risk factors for heart attack include:

1. **Unhealthy Diet**: A diet high in saturated fats, trans fats, and cholesterol can contribute to heart disease.
2. **Physical Inactivity**: Lack of regular physical activity increases the risk of heart disease.
3. **Tobacco Use**: Smoking and other forms of tobacco use significantly raise the risk of heart attacks.
4. **Harmful Use of Alcohol**: Excessive alcohol consumption can lead to high blood pressure and other health issues.
5. **Raised Blood Pressure**: High blood pressure is a major risk factor for heart disease.
6. **Raised Blood Glucose**: High blood sugar levels can increase the risk of heart disease, especially in individuals with diabetes.
7. **Raised Blood Lipids**: High levels of cholesterol and triglycerides in the blood can lead to heart disease.
8. **Overweight and Obesity**: Excess body weight increases the risk of heart disease and related conditions.

These risk factors can be measured in primary care facilities and indicate an increased risk of heart attack, stroke, heart failure, and other complications (Cardiovascular diseases.pdf, Page 1).
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8. Animal Bites

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User: cobra bite my cousin, what to do now

Healthcare Assistant: If someone has been bitten by a cobra, it is crucial to seek prompt medical attention. The main steps of care include:

1. **Call for emergency help** immediately to transport the person to a medical facility.
2. **Keep the person calm and still** to slow the spread of venom.
3. **Remove any tight clothing or jewelry** near the bite site to allow for swelling.
4. **Do not apply ice or a tourniquet**, and do not attempt to suck out the venom.
5. **Monitor for any signs of breathing difficulties, acute allergic reactions, or anaphylaxis**.

Prompt medical attention is key, and treatment may include the administration of appropriate antivenom (Animal_bites.pdf, Page 1).
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```

9. Bot not answering the questions that are out of documents that were ingested.

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User: how to overcome diabetes?

Healthcare Assistant: I'm not sure based on the available healthcare documents.
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User: can you tell me the latest news by PM modi?

Healthcare Assistant: I'm not sure based on the available healthcare documents.
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User: is world leading to World war 3?

Healthcare Assistant: I'm not sure based on the available healthcare documents.
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User: whats your opinion on varanasi movie by Rajamouli

Healthcare Assistant: I'm not sure based on the available healthcare documents.
-----
User: exit
labsuser@jupyter:~/work/RAG Project$ █
```